

SignSpeak

Project Description

SignSpeak is an AI-powered sign language interpreter tool designed to support communication during video calls. The web-based application translates American Sign Language (ASL) gestures into text and/or spoken voice through a simple webcam. Its mission is to bridge communication barriers between deaf and hearing communities by offering an intuitive, accessible, and hardware-free solution that fosters inclusive conversation.

Value Proposition

For deaf, hard-of-hearing individuals, and hearing people who want to communicate with them without knowing sign language, who lack accessible tools for fluid and private conversations during online meetings and video calls, SignSpeak is an AI-powered web application that enables communication without the need for a human interpreter, promoting user independence and privacy. Unlike human interpretation services, complex hardware systems (like SignAll), or apps that only translate text-to-sign (like HandTalk), our product uses AI-based gesture recognition through a simple webcam to translate sign language (ASL) into text or speech during video calls, making it more accessible and affordable for everyday use.

Team Members, Leader, and Preliminary Roles

- **Karol Jinet Cardona Bolanos : Product owner /Fontend Developer/ Tester Engineer** - Defines vision, manages backlog, ensures alignment with user needs. Tests functionality, latency, and accessibility across devices and browsers.
- **Elena Martinez Vázquez : UX/UI Designer / Frontend Developer** - Builds the web interface , integrates webcam capture, ensures accessibility.
- **Michal Aleksander Sachanbinski : Machine Learning Engineer** - Develops and optimizes ASL recognition models.
- **Mohamed Salah Abdelaziz Mohamed Heikal : Product Manager / Machine Learning Engineer** – Develops and optimizes ASL recognition models.
- **Kuessi-Ansan Ansan Manuel DJIYEHOUÉ : Backend Developer** - Manages the **WebSocket** infrastructure that synchronizes state and audio triggers between Signers and Listeners in real-time.

User Personas

Persona 1:

- **Name:** Rebecca - Deaf University Student
- **Demographics:** 21, ASL native signer, Communication major.
- **Goals:**

- Communicate with professors and peers without needing an interpreter.
- Participate actively in group discussions and debates.
- Access lectures and workshops spontaneously without prior scheduling.
- **Frustrations:**
 - Scheduling interpreters is slow
 - Video group discussions are often inaccessible.
 - Many online platforms lack integrated accessibility tools.
- **Quote:** *"I just want to join any video class without planning days in advance or depending on a human translator."*

Persona 2:

Name: David – Hearing Participant on a Call

- **Demographics:** 22, hearing student, Rebecca's classmate.
- **Goals:**
 - Understand Rebecca's ideas and questions.
 - Have a productive and inclusive group discussion.
 - Not interrupt the flow of the conversation by waiting for Rebecca to type.
- **Frustrations:**
 - Current group discussions are slow and feel fragmented.
 - Fears he is missing Rebecca's valuable contributions because communication is so inefficient.
- **Quote:** *"I know Rebecca has great ideas, but the conversation always stops while we wait for her to type. I just wish we could understand her instantly."*

Persona 3:

Name: Lucy – Parent of a Deaf Son

- **Demographics:** 58, basic ASL learner. Her son became deaf few months ago.
- **Goals:**
 - To have a natural conversation without constantly stopping to ask "Wait, what did you say?" or asking him to slow down.
 - To understand complex details (like medical appointments or work issues) accurately, without relying on guesswork.
 - To keep eye contact with her son as much as possible, glancing at the translation only when she misses a sign.
- **Frustrations:**
 - Feels guilty when she has to interrupt her son's exciting stories because she lost track of the signing.
 - Finds that typing on a phone to clear up misunderstandings kills the mood and spontaneity of the moment.
 - Is afraid of misunderstanding important information because her ASL vocabulary is limited to basic topics.
- **Quote:** *"I love hearing his stories, but sometimes he gets so fast that I get lost. I just want a tool that helps me catch the words I miss, so we can keep talking like normal."*

User Stories

Rebecca (Student)

- As a deaf student, I want to join a session as a **Signer**, so my camera is automatically activated to capture my ASL gestures.
- As a user, I want to detach the translation window using Picture-in-Picture, so I can place the subtitles right next to my professor's face on Zoom without hiding the lecture slides
- As a presenter, I want a Restart/Clear button that allows me to instantly wipe the current translation buffer if I notice the AI misunderstood my gesture, so I can re-sign it immediately without sending incorrect text to my classmates.
- As a learner, I want the tool to work without extra hardware, so I can use it anywhere on campus.
- As a user, I want to use the Meeting ID to sync instantly with the hearing people in the lecture call, in this way they can read or listen to my translation.

David (Hearing Participant on a Call)

- As a hearing participant in a video call, I want to join the session as a **Listener** (without sending my own video), so I can receive the translation of my deaf colleague directly on my screen.
- As a participant who prefers auditory learning, I want to click a 'Speaker' button to hear the translation spoken aloud (Text-to-Speech), allowing me to listen to my classmate's question without looking away from the professor or the slides.
- As a hearing classmate, I want the translated text to appear word-by-word as my colleague signs, so the conversation feels natural and doesn't have an awkward delay.
- As a user, I want to use the Meeting ID to sync instantly with the deaf colleague's session.

Lucy (Parent)

- As a mother, I want the application to work in **Local Mode** to verify my understanding of his signing in real-time, so I don't have to interrupt him constantly to ask for clarification.
- As a user with limited ASL vocabulary, I want the tool to translate complex signs into text instantly, giving me the confidence that I am understanding the full context of the conversation.
- As a user with basic tech skills, I want the tool to be simple and reliable, so I can start a conversation with my son instantly without worrying about complex configuration or setup.
- As a user, I want to be able to increase the font size, ensuring the text is readable on my screen during the call.